

PHASE 3 — Attendance Management

PRSECURITY HRMS

IMPORTANT RULES

Output every file 100% completely. Never use '...' or 'rest remains same'.

After each file write "--- FILE COMPLETE ---" then start the next file immediately.

If about to hit output limit, finish current file and write "--- PAUSED: Type CONTINUE ---"

This is Phase 3 of 8. Phases 1 & 2 already built: Auth, Employees, Dashboards.

DO NOT rebuild previous phase files. Only add NEW files or EXTEND existing ones where specified.

CONTEXT

Project: PRSECURITY HRMS | Stack: React+Vite+TS+Tailwind+shadcn/ui |

Node+Express+Prisma+PostgreSQL Roles: ADMIN, HR, EMPLOYEE, INTERN | Currency: ₹ | Dates: DD

MMM YYYY Attendance: Hybrid office — capture geolocation + IP on punch. No hard geofencing, just record + display.

STEP 1 — Update Prisma Schema: Attendance Model

Replace the placeholder Attendance model with the full schema:



prisma

```

model Attendance {
  id      String @id @default(uuid())
  userId  String
  date    DateTime @db.Date
  punchIn DateTime?
  punchOut DateTime?
  punchInLat Float?
  punchInLong Float?
  punchOutLat Float?
  punchOutLong Float?
  punchInIP String?
  punchOutIP String?
  workingHours Float?
  status  AttendanceStatus @default(ABSENT)
  notes   String?
  isManualEntry Boolean @default(false)
  approvedBy String?
  approver User? @relation("AttendanceApprover", fields: [approvedBy], references: [id])
  user    User @relation(fields: [userId], references: [id])
  createdAt DateTime @default(now())
  updatedAt DateTime @updatedAt

  @@unique([userId, date])
}

enum AttendanceStatus {
  PRESENT
  ABSENT
  HALF_DAY
  HOLIDAY
  WEEKEND
  ON_LEAVE
}

```

Run: `npx prisma migrate dev --name add_attendance`

STEP 2 — Attendance Backend

FILE: backend/src/services/attendance.service.ts

```
punchIn(userId, lat?, long?, ip):
```

Check no punch-in exists for today

Create Attendance record with date=today, punchIn=now, punchInLat, punchInLong, punchInIP

Set status = PRESENT

Return record

```
punchOut(userId, lat?, long?, ip):
```

Find today's attendance record for user

Throw error if no punch-in found

Throw error if already punched out

Update: punchOut=now, punchOutLat, punchOutLong, punchOutIP

Calculate workingHours = (punchOut - punchIn) in decimal hours (e.g. 7.5)

If workingHours < 4: set status = HALF_DAY

Return updated record

```
getTodayAttendance(userId):
```

Return today's attendance record for user or null

```
getMyAttendance(userId, month, year):
```

Return all attendance records for user for given month/year

Include weekends as WEEKEND status and holidays from HolidayCalendar

```
getAllAttendance(query):
```

Paginated list: all users' attendance for a given date or date range

Params: date (default today), userId (optional), status, page, limit

Include user name and employeeId

```
getMonthlyAttendanceSummary(userId, month, year):
```

Count: workingDays (excluding weekends+holidays), presentDays, absentDays, halfDays, leaveDays

Calculate attendance percentage

Return summary object

```
manualAttendanceEntry(data, adminUserId):
```

HR/Admin creates or updates an attendance record for any user

Set isManualEntry=true, approvedBy=adminUserId

```
getAttendanceStats():
```

For today: total present, total absent, total on leave

Return counts

FILE: backend/src/controllers/attendance.controller.ts

Controllers for all service functions. Extract IP from req.ip or req.headers['x-forwarded-for'].

FILE: backend/src/routes/attendance.routes.ts

```
□  
  
POST /api/attendance/punch-in    → punchIn (EMPLOYEE/INTERN/HR/ADMIN)  
POST /api/attendance/punch-out  → punchOut (EMPLOYEE/INTERN/HR/ADMIN)  
GET  /api/attendance/today      → getTodayAttendance (authenticated)  
GET  /api/attendance/my         → getMyAttendance?month=&year= (authenticated)  
GET  /api/attendance/my/summary → getMonthlyAttendanceSummary?month=&year= (authenticated)  
GET  /api/attendance            → getAllAttendance (HR/ADMIN)  
GET  /api/attendance/:userId     → getUserAttendance (HR/ADMIN)  
POST /api/attendance/manual     → manualAttendanceEntry (HR/ADMIN)  
GET  /api/attendance/stats/today → getAttendanceStats (HR/ADMIN)
```

STEP 3 — Attendance Frontend

FILE: `frontend/src/api/attendance.api.ts`

Functions for all attendance endpoints.

FILE: `frontend/src/components/attendance/PunchWidget.tsx`

Large punch in/out widget used on employee dashboard:

Shows current time (updates every second)

Shows today's date

Big button: "Punch In" (green) when not punched in, "Punch Out" (red) when punched in

On click: request browser geolocation (`navigator.geolocation.getCurrentPosition`)

If granted: send lat/long with request

If denied: send request without lat/long (still works)

Shows punch-in time after punching in

Shows punch-out time + working hours after punching out

Shows "Already completed for today" if both punched

Loading state on button while API call in progress

FILE: `frontend/src/components/attendance/AttendanceCalendar.tsx`

Monthly calendar heatmap component: Props: `{ records: AttendanceRecord[], month: number, year: number, onMonthChange: fn }`

Grid of days (7 columns)

Colour per status: Present=green, Absent=red, Half Day=yellow, Leave=blue, Weekend=light grey, Holiday=purple

Each day shows status initial (P/A/H/L/W/Ho)

Hover tooltip: full date + status + punch times if available

Previous/Next month navigation

Legend at bottom

FILE: `frontend/src/components/attendance/AttendanceSummaryCard.tsx`

Component showing monthly summary:

Working Days, Present, Absent, Half Days, On Leave, Attendance %

Circular progress showing attendance percentage

FILE: `frontend/src/pages/attendance/AttendancePage.tsx`

Employee/Intern view:

Month/Year selector at top right

PunchWidget (today's punch)

AttendanceSummaryCard for selected month

AttendanceCalendar for selected month

Attendance Log table below calendar:

Columns: Date, Day, Punch In, Punch Out, Working Hours, Status, Location (shows " 📍 Captured" if lat/long exists, clickable to show coordinates)

Paginated, 31 rows max per month

FILE: `frontend/src/pages/attendance/TeamAttendancePage.tsx`

HR/Admin view — 2 tabs:

Tab 1 — Today's Overview:

Stats row: Present, Absent, On Leave (using `attendance/stats/today` endpoint)

Table: all employees, their today status, punch-in time, punch-out time, working hours

Status badge per row

Search by name

Tab 2 — Monthly Logs:

Month/Year selector

DataTable: Employee Name, Employee ID, Total Present, Total Absent, Half Days, Leave Days, Attendance %

Click row → opens employee's detailed monthly log in a modal/drawer

Export CSV

FILE: frontend/src/components/attendance/ManualEntryModal.tsx

HR/Admin modal to manually add/edit an attendance record:

Select Employee (dropdown), Date picker

Status dropdown, Punch In time, Punch Out time

Notes field

Submit → calls manual entry API

Add "Manual Entry" button to TeamAttendancePage that opens this modal.

STEP 4 — Connect Punch Widget to Employee Dashboard

Update frontend/src/pages/dashboard/EmployeeDashboardPage.tsx

Replace the placeholder punch widget with the real `<PunchWidget />` component

Below it show `<AttendanceSummaryCard />` for current month

Fetch today's attendance on mount and pass to PunchWidget

STEP 5 — Update Employee Detail Page (from Phase 2)

Update frontend/src/pages/employees/EmployeeDetailPage.tsx

Replace the "Attendance" tab EmptyState with:

Month/Year selector

AttendanceSummaryCard

AttendanceCalendar

Attendance log table

STEP 6 — Update Admin Dashboard Stats

Update backend/src/services/dashboard.service.ts

Fill in the `presentToday` and `onLeaveToday` counts using real Attendance data.

AFTER ALL FILES ARE COMPLETE

Write this exactly: "  PHASE 3 COMPLETE — Attendance punch in/out with geolocation capture, monthly calendar view, team attendance overview, and manual entry are ready. Proceed to Phase 4 for Leave Management."